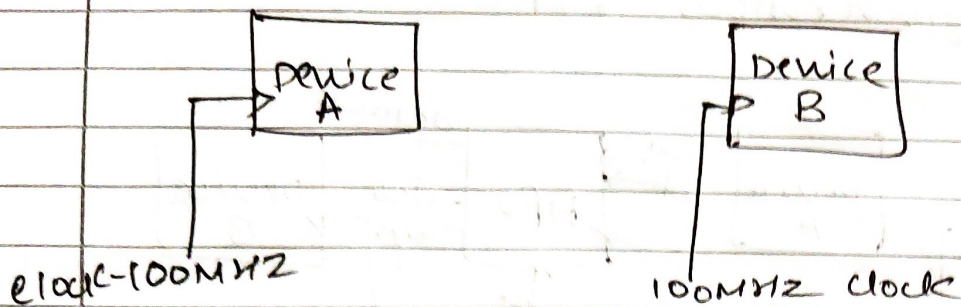
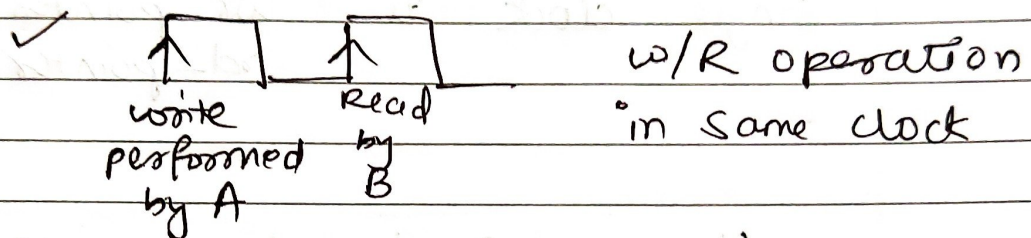


Prerna Agrewal Asynchronous fifo (data Buffer)



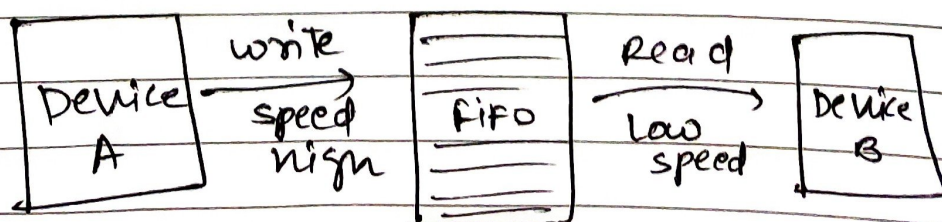
- ✓ same clock frequency
- ✓ synchronous



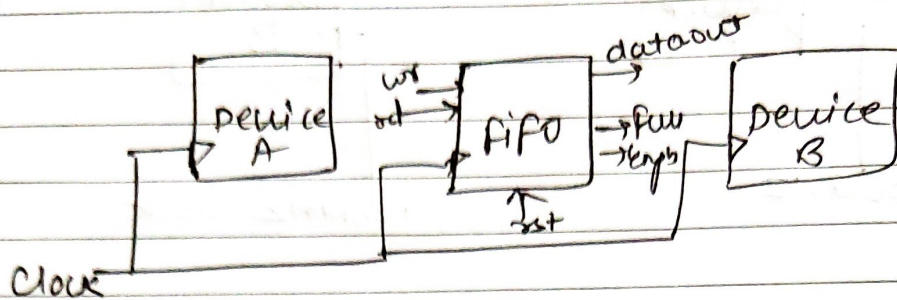
- ✓ writing speed = Reading speed

जिनका data device A ने write करत
उसका data device B ने Read करत

→ When writing rate $\neq$ reading rate
meaning some data has not been
read by device. ($\therefore$ Data corruption)
hence in this case, we use
FIFO to store all data



if synchronous FIFO

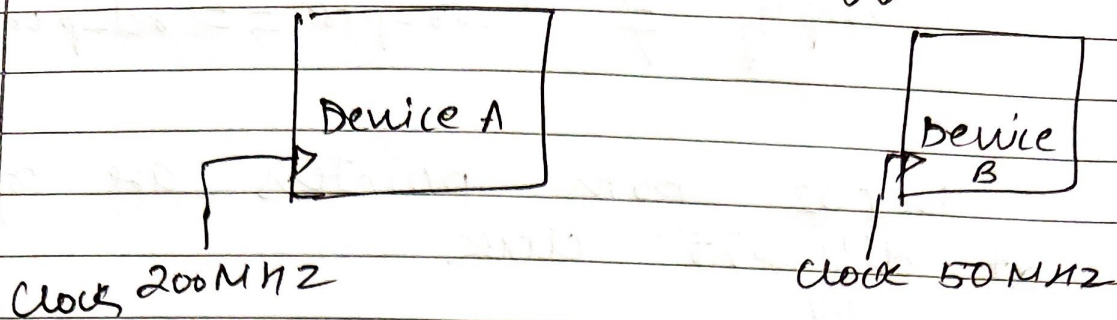


FIFO wr/rd operation perform in
Single clock & wr-pointer } increments
rd-pointer } in same
clock

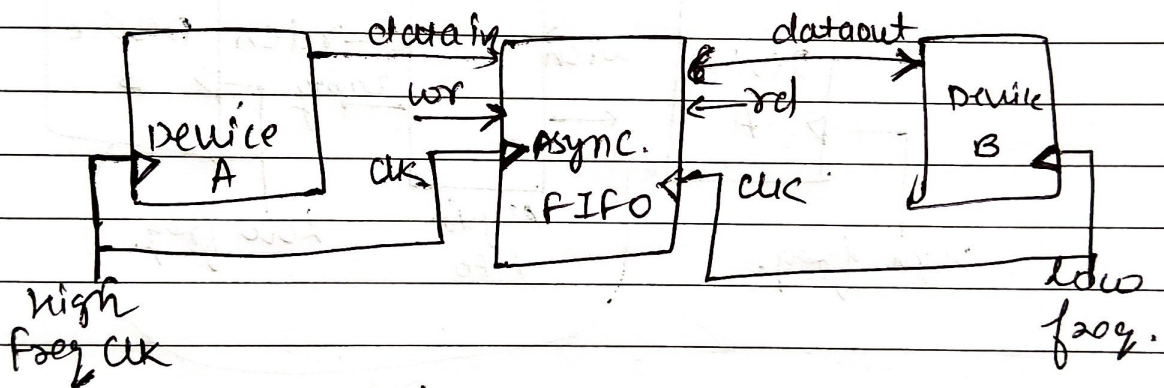
As all Device A, Device B, FIFO
~~operates~~ in same clock. So there is
driven by no problem in functionality.

//_

Problem occurs when Both devices are operating with different clocks



we need to select at ~~what~~ which clock should we perform write operation ~~or~~ read operation.



Here, write operation will happen w.r.t High freq clock

Read operation will be performed w.r.t low freq. clock.

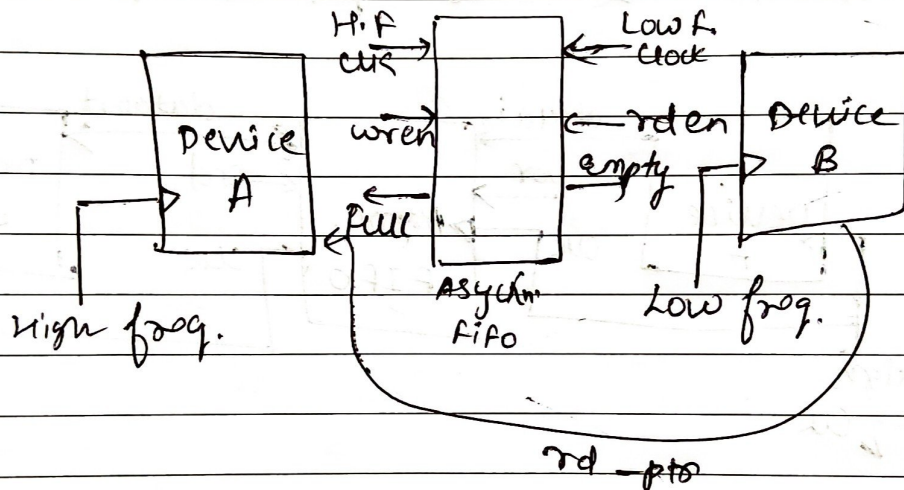
Now
wr_ptr $\xrightarrow{\text{incr.}}$ at high clock freq.
rd_ptr $\xrightarrow{\text{incr.}}$ at low clock freq.

But in Asynchronous fifo

empty $\neq$ (wr-pts == rd-pts)

Beoz Both pointers are running at different clock.

Q So how do we calculate full & empty condition in Asynchronous fifo?



rd-pts increment which is happening at low frequency domain should be shared to high freq. domain
Beoz wr-pts should be compared
rd-pts
for ^{evaluating} "full condition"

Similarly, wr-pts is shared to low freq. domain for comparing rd-wtr
so to evaluate "empty" flag

22/8/26
DATE

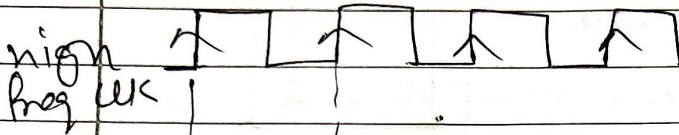
* wr_ptr $\Rightarrow$ Binary value

rd_ptr $\Rightarrow$ Binary value

(High freq)

(Low freq)

Shared



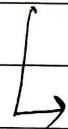
rd_ptr value $\rightarrow$ 000 001

000 If setup violation then
00x metastability state

001 after some time settled

010
011

0xx again metastability state occurs

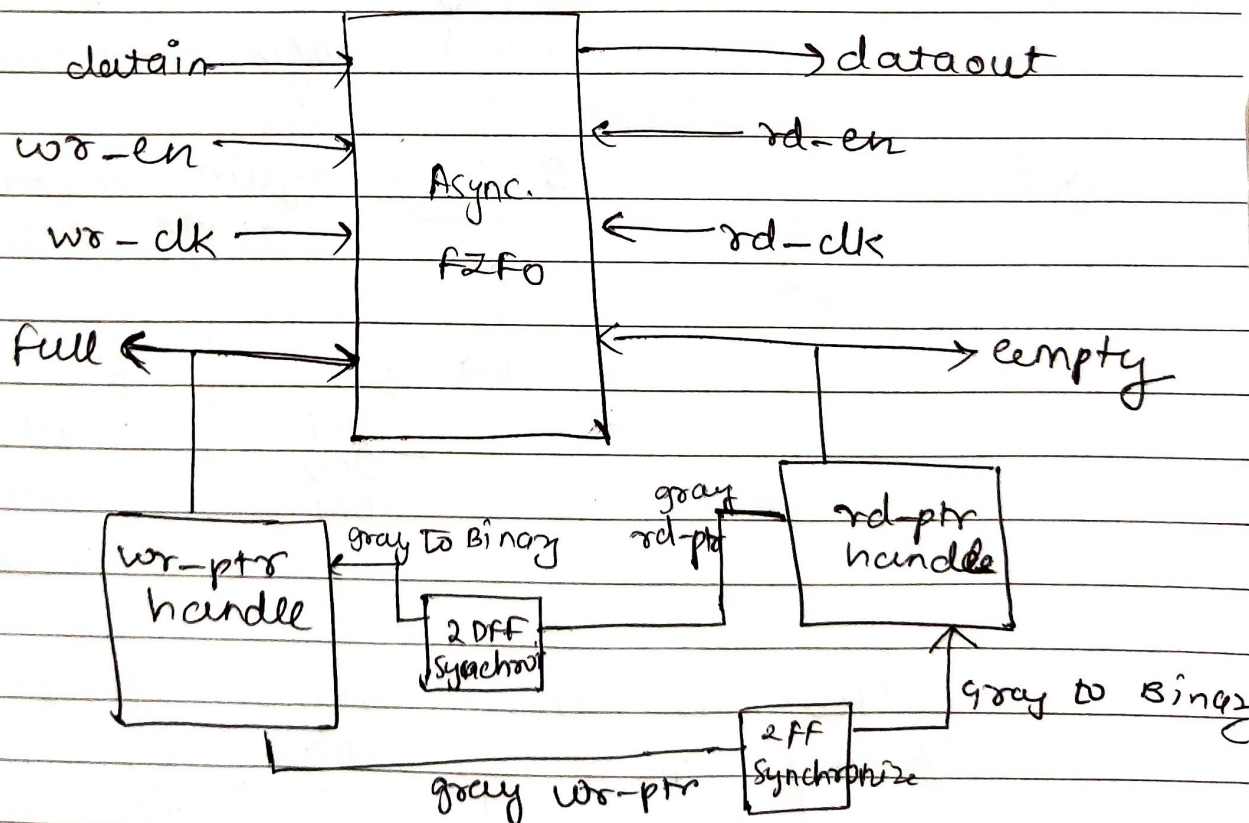
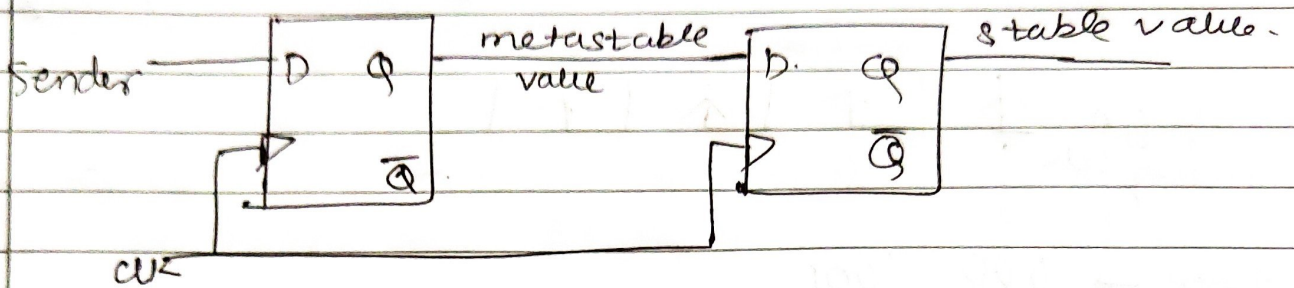


000
001
010
011 } after some time
rd_ptr value
can be any of
these values

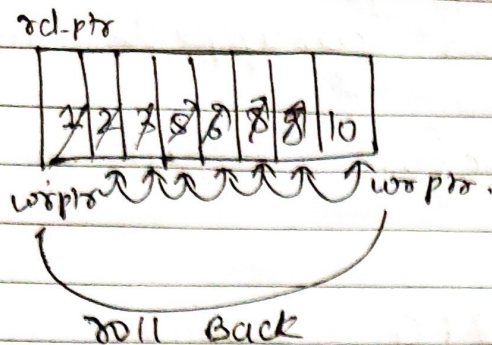
This is the major problem with Binary pointers. Therefore, Gray pointers are used in Asynchronous FIFO.

metastability might be occur in Gray code pointers as well But we get stable single value.

Now to Eliminate Metastability
if Gray coding is used for pointers
we use 2-DFF Synchronizers



How to Evaluate Full & Empty Condⁿ?



FIFO is full

But wr_ptr rolls back to 0 location

Both rd_ptr & wr_ptr are at same location

wr_ptr == rd_ptr → empty

So Full or Empty ??

To avoid this confusion we add extra Bit

eg:-

FIFO depth = 8

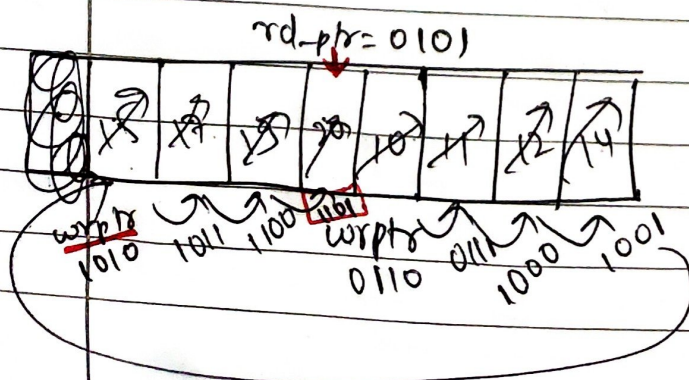
wr_ptr = $\log_2(\text{depth}) = 3\text{bit}$

rd_ptr = $\log_2(\text{depth}) = 3\text{bit}$

Now 2 Extra Bit Added to both pointers

$\therefore \log_2(\text{depth}) + 1$ wide.

pointer width = 4 bits



→ All 8 locations are written

wr_ptr = 1101

rd_ptr = 0101

//_

MSB Bit differs and indicates that
wr-ptr has wrapped more than
Rd-ptr. Therefore, full is detected

Note

Depth of FZFO remains same.

Fifo Depth

__/__/__

Case
1

$$F_{write} = 60 \text{ MHz}$$

$$F_{read} = 30 \text{ MHz}$$

No. of data items transferred
(Burst) length = 100

→ writing rate is faster than reading rate
 $F_{write} > F_{read}$

$$\begin{aligned} \text{Time taken to write} &= \frac{1}{60 \times 10^6} \\ \text{1 data item} & \\ &= 16.66 \text{ nsec} \end{aligned}$$

$$\begin{aligned} \text{Time for writing 100 data items} & \\ &= 16.66 \times 100 \\ &= 1660 \text{ nsec} \end{aligned}$$

$$\text{Time to read 1 data item} = \frac{1}{30 \times 10^6} = 33.3 \text{ nsec}$$

if 1 data item $\xrightarrow{\text{read}}$ 33.3 nsec
? $\xrightarrow{\text{read}}$ 1660 nsec

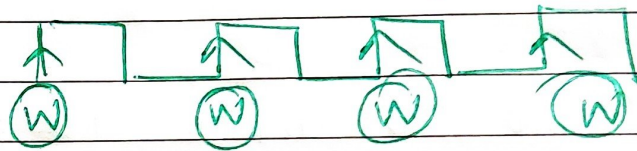
$$n = \frac{1660 \text{ nsec}}{33.3 \text{ nsec}} = 49.8 \approx 50 \text{ data items}$$

$$100 - 50 = 50 \text{ data items are left to read out}$$

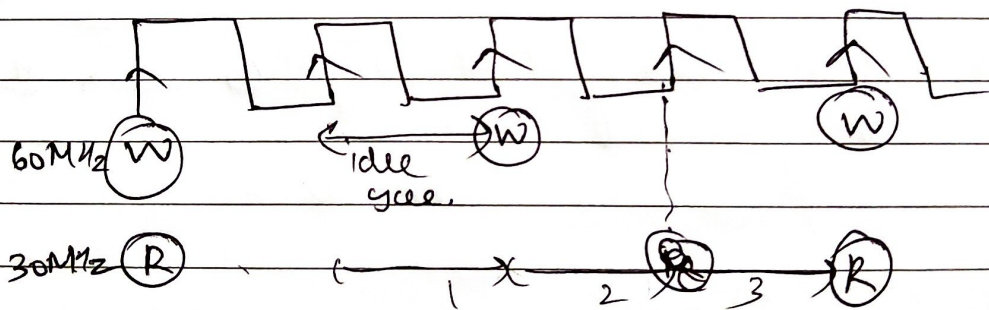
remaining

$$\therefore \text{depth of FIFO} = 50$$

In this case we assumed that write operation is performed cycle by cycle (No idle cycles) present



Case-2



No. of idle cycles b/w consecutive write = 1

No. of idle cycles b/w consecutive read = 3

Data items $\Rightarrow 100$

Meaning, after writing 1 data device A is waiting for 1 clock cycle to initiate next write.

//_

So we can understand that
for every 2 cycles $\rightarrow$ 1 write is performed

$$T_{\text{write}} = \frac{1}{60 \times 10^6} = 16.66 \text{ nsec}$$

2 cycles.

$$\text{Time taken to } \underset{\text{write}}{\text{write}} \text{ 1 data item} = 2 \times 16.666 = 33.32 \text{ nsec}$$

$$\text{Time to write 100 data items} = 100 \times 33.32 = 3332 \text{ nsec}$$

$\rightarrow$ for every 4 cycles $\rightarrow$ 1 read operation is performed

$$T_{\text{read}} = \frac{1}{30 \times 10^6} = 33.33 \text{ nsec}$$

$$\text{Total time to read 1 data item} = 4 \times 33.33 = 133.32 \text{ nsec}$$

$$1 \text{ data item } \xrightarrow{\text{read}} 133.32 \text{ nsec}$$

$$\times \xrightarrow{\text{read}} 3332 \text{ nsec}$$

$$n = \frac{3332}{133.32} = 25 \text{ data items}$$

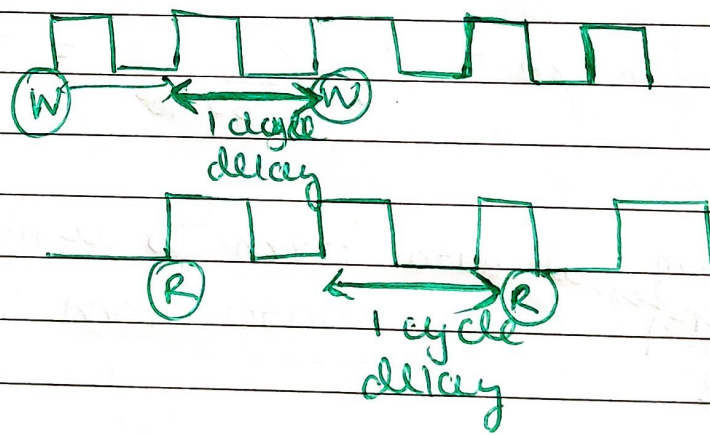
$$100 - 25 = 75$$

/ Total written data item \ read
 = data items are left to be read so the size of fifo

depth of fifo = 75

so that fifo is capable of storing 75 data items.

Case-3 if A > B with 1 clock cycle delay between 2 successive reads & writes.



it is just creating sort of confusion.

for every 2 cycles → 1 write is performed

for every 2 cycles → 1 Read is performed

so the case is same as previous case

Case 4 $f_A > f_B$ with duty cycles

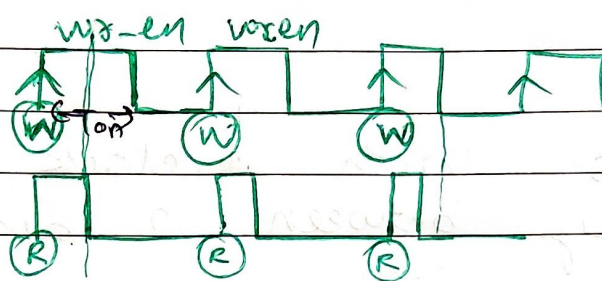
(f_A) Write frequency = 80 MHz

(f_B) Read frequency = 50 MHz

Burst length = 120

Duty Cycle of write $\rightarrow 50\%$

Duty cycle of read $\rightarrow 25\%$



$$\begin{aligned} (T_{on})_{wr} &= \frac{1}{2} \times (T_{wr}) \\ &= \frac{1}{2} \times \left(\frac{1}{80 \times 10^6} \right) \\ &= 6.25 \text{ ns} \end{aligned}$$

$$(T_{on})_{rd} = \frac{1}{4} \times \left(\frac{1}{50 \times 10^6} \right) = 5 \text{ ns}$$

1 data item is not transferred in 6.25 ns

1 data item to write $\rightarrow 6.25 \text{ ns}$

120 data item $\rightarrow 120 \times 6.25 \text{ ns}$
750 ns

1 data item to be read $\rightarrow 5 \text{ ns}$

No. of data items to be read in 750 ns
 $= \frac{750}{5}$
 $= 150$

Write enable is active $\rightarrow 0.5$ cycle

Since 120 transfers require 120 enabled cycles

//_

only 50% of cycles are enabled $= \frac{120}{0.5} = 240$ cycles

$$240 \times (12.5ns) = 3\mu s$$

$$\text{Data read out} = 3\mu s \times (50MHz \times \frac{1}{4}) = 37 \text{ items}$$
$$\text{Fifo depth} = 120 - 37 = 83$$

Case-5

$$f_{\text{req write}} = 30MHz$$

$$f_{\text{req read}} = 50MHz$$

No idle cycles in Read & write means all data items in the Burst will be written and reads in consecutive clock cycles

→ Reading rate > writing rate

→ Fifo depth is 1 to prevent any data loss as Reading is faster than writing

Case-6

$$f_{\text{req write}} = 30MHz \quad f_{\text{req read}} = 50MHz$$

Burst length = 120 no. of data items to be transferred

No. of Idle cycles b/w 2 successive write = 1

No. of Idle cycle b/w 2 successive read = 3

for every 2 clock cycles $\rightarrow$ 1 data is written

$$\begin{aligned}\text{Time to write 1 data item} &= 2 \times \frac{1}{30 \text{ MHz}} \\ &= 66.666 \text{ ns}\end{aligned}$$

$$\begin{aligned}\text{Time to write 120 data items} &= 66.666 \times 120 \\ &= 8000 \text{ ns}\end{aligned}$$

for every 4 clock cycles $\rightarrow$ 1 data is read

$$\begin{aligned}\text{Time to read 1 data item} &= 4 \times \frac{1}{50 \times 10^6} \\ &= 80 \text{ ns}\end{aligned}$$

8000 ns $\xrightarrow{\text{read}}$ 1 data item

$$1 \text{ ns} \rightarrow \frac{1}{80 \text{ ns}}$$

$$8000 \text{ ns} \rightarrow \frac{8000}{80} = 100$$

$$\begin{aligned}\text{remaining bytes to be} \\ \text{left for read} \quad \text{But} &= 120 - 100 \\ &= 20\end{aligned}$$

$$\text{depth of fifo} = 20$$

//_

Case-7 $f_{write} = f_{read}$
 $\downarrow \qquad \qquad \downarrow$
 $30\text{ MHz} \qquad 30\text{ MHz}$

Burst length = 120

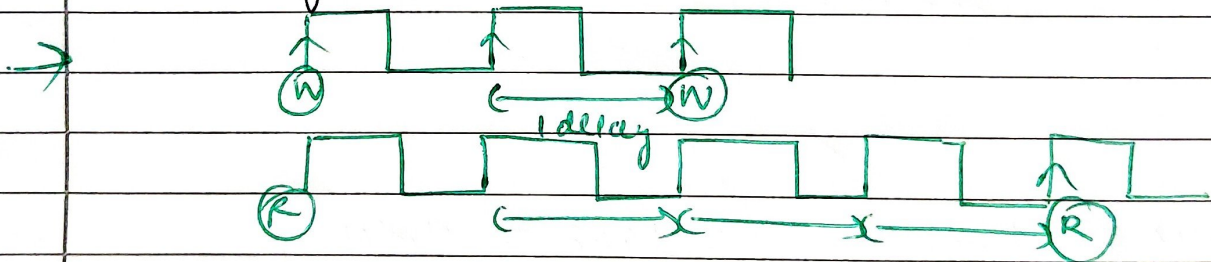
→ Fifo is not required if there is no phase difference between clk_A & clk_B

Case-8 $f_{write} = 50\text{ MHz}$ $f_{read} = 50\text{ MHz}$

Burst length = 120

No. of idle cycles b/w 2 successive write is 1

No. of idle cycles b/w 2 Read is 3



→ Time required to write 1 data item
 $= 2 \times \left(\frac{1}{50\text{ MHz}} \right)$
 $= 40\text{ ns}$

→ Time required To write 120 data item
 $= 120 \times 40\text{ ns}$
 $= 4800\text{ ns}$

//_

$$\begin{aligned}\text{Time required to read 1 data item} \\ &= 4 \times \left(\frac{1}{50 \text{ MHz}} \right) \\ &= 80 \text{ ns}\end{aligned}$$

? data items to read in 4800 ns

$$n = \frac{4800}{80} = 60$$

Remaining data items = $120 - 60$
to be stored in FIFO = 60

$$\begin{aligned}\text{Time required to read 1 data item} \\ &= 4 \times \left(\frac{1}{50 \text{ MHz}} \right) \\ &= 80 \text{ ns}\end{aligned}$$

? data items to read in 4800 ns

$$n = \frac{4800}{80} = 60$$

$$\begin{aligned}\text{Remaining data items} &= 120 - 60 \\ \text{to be stored in FIFO} &= 60\end{aligned}$$

8 $f_{\text{write}} = 200 \text{ MHz}$
 write_en is high for 3 clock cycles
 Low for 1 clock

$f_{\text{read}} = 300 \text{ MHz}$
 read_en is high for 7 clock
 Low for 8 clk.

a) Eff. Writing rate?
 Effective Reading rate?

$$\text{Total write clocks} = 4 \text{ (High + Low)}$$

$$\begin{aligned}\rightarrow \text{Effective writing rate} &= \frac{3}{4} \times 200 \text{ MHz} \\ &= 150 \text{ MHz}\end{aligned}$$

$$\rightarrow \text{Effective reading rate} = \frac{7}{15} \times 300 = 140 \text{ MHz}$$

//_

Now writing rate $>$ Reading rate

b) if Depth of FZFO = 200
Then Total no. of data written
into FZFO before it is full.

→

$$\begin{aligned}\text{Net rate} &= 150 \text{ Million words/sec} - 140 \text{ million words/sec} \\ &= 10 \text{ Million words/sec}\end{aligned}$$

$$\begin{aligned}\text{Time to fill FZFO depth of } 200 &= \\ \frac{200}{10 \times 10^6} &= 20 \mu\text{s}\end{aligned}$$

$$\begin{aligned}\text{Total Active write in } 20 \mu\text{s} &= 150 \text{ M words/sec} \times 20 \mu\text{s} \\ &= 3000 \text{ cycles}\end{aligned}$$

$$\begin{aligned}\text{Total Active read in } 20 \mu\text{s} &= 140 \text{ M words/sec} \times 20 \mu\text{s} \\ &= 2800 \text{ cycles}\end{aligned}$$